Battery Life Predictor Based on Duty Cycle Model

1. Aim

To design and implement a simple battery life prediction system based on the duty cycle model, which estimates the operational lifetime of a battery-powered embedded system using active and sleep current consumption patterns.

2. Theory

In embedded systems, power consumption is a critical factor, especially for battery-operated devices such as IoT sensors, wearables, and remote monitoring systems. These systems typically operate in two modes:

- **Active Mode:** The system performs tasks like sensing, processing, or communication, consuming higher current.
- **Sleep Mode:** The system remains idle or in low-power state, consuming minimal current.

The duty cycle represents the fraction of time the system stays in active mode:

Duty Cycle = Active Time / (Active Time + Sleep Time)

To estimate battery life, we calculate the average current consumption:

Average Current (I_avg) =(Duty Cycle × Active Current) + ((1 − Duty Cycle) × Sleep Current)

Finally, battery life is computed as:

Battery Life (in hours) = Battery Capacity / Average Current

Assumptions

- Battery discharge is linear.
- No losses due to temperature, aging, or internal resistance.
- Constant current consumption in active and sleep modes.

3. System Architecture

The system is a **software-based model** that simulates battery usage using user-provided parameters.

Inputs:

- Battery Capacity (mAh)
- Active Current Consumption (mA)
- Sleep Current Consumption (mA)
- Active Time (seconds)
- Sleep Time (seconds)

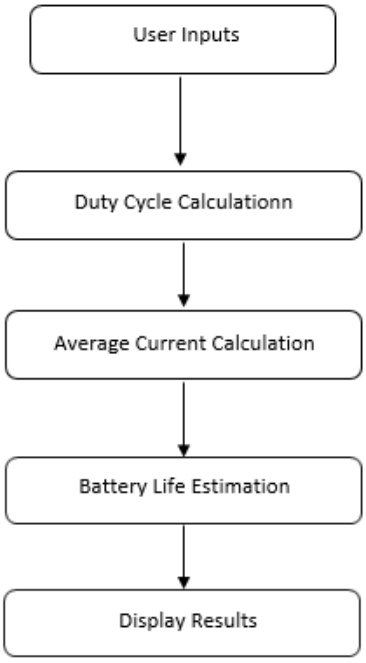
Processing Module

This module performs all necessary calculations using the duty cycle model:

1. **Computes total time:**
   Total Time = Active Time + Sleep Time
2. **Calculates duty cycle:**
   Duty Cycle = Active Time / Total Tim
3. **Calculates average current:**
   Average Current = (Duty Cycle × Active Current) + ((1 − Duty Cycle) × Sleep Current)
4. **Estimates battery life:**
   Battery Life = Battery Capacity / Average Current

This module acts as the core logic of the system.

4. Block Diagram:



5. Code:

```
def battery_life_predictor():
    print("=== Battery Life Predictor (Duty Cycle Model) ===")

    capacity = float(input("Enter Battery Capacity (mAh): "))
    I_active = float(input("Enter Active Current (mA): "))
    I_sleep = float(input("Enter Sleep Current (mA): "))
    T_active = float(input("Enter Active Time (seconds): "))
    T_sleep = float(input("Enter Sleep Time (seconds): "))

    T_total = T_active + T_sleep
    duty_cycle = T_active / T_total

    I_avg = duty_cycle * I_active + (1 - duty_cycle) * I_sleep
    battery_life = capacity / I_avg
```

```
    print("\n--- Results ---")
    print(f"Duty Cycle: {duty_cycle:.3f} ({duty_cycle*100:.2f}%)")
    print(f"Average Current: {I_avg:.2f} mA")
    print(f"Estimated Battery Life: {battery_life:.2f} hours")

battery_life_predictor()
```

6. Input values:

Battery Capacity = **1500 mAh**

Active Current = **80 mA**

Sleep Current = **5 mA**

Active Time = **8 seconds**

7. Outputs:

- Duty Cycle (%)
- Average Current Consumption (mA)
- Estimated Battery Life (hours)

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS 1

@likhithakruthi ➜/workspaces/embedded (main) $ cd assignment/
@likhithakruthi ➜/workspaces/embedded/assignment (main) $ python duty_cycle.py
=== Battery Life Predictor (Duty Cycle Model) ===
Enter Battery Capacity (mAh): 1500
Enter Active Current (mA): 80
Enter Sleep Current (mA): 5
Enter Active Time (seconds): 8
Enter Sleep Time (seconds): 2

--- Results ---
Duty Cycle: 0.800 (80.00%)
Average Current: 65.00 mA
Estimated Battery Life: 23.08 hours
@likhithakruthi ➜/workspaces/embedded/assignment (main) $
```

8. Conclusion:

The implemented model provides a simple and effective way to estimate battery life based on duty cycle. While it is suitable for initial design analysis, real-world applications may require more complex models that consider environmental and hardware-specific factors.